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A multi-agent system framework for manufacturing planning and control*

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***Abstract* - This paper presents a hybrid multi-agent system framework for manufacturing planning and control activities. The hybrid framework comprises a supervisor agent and function agents to incorporate both decentralized negotiation and hierarchical control. A hybrid-based agent structure is introduced in the design of individual agents. Each agent consists of a planning layer and a reactive layer, to cope with the planning and control functions respectively. Manufacturing process plans and schedule are to be determined through agent negotiations. Accordingly, a multi-agent negotiation protocol is designed for effectively coordinating the interactions between the planning layers of the supervisor agent and the local agents.**

Index Terms – multi-agent systems, manufacturing planning and control, process plan, production schedule

I. INTRODUCTION

Manufacturing systems are commonly established with the conventional hierarchical control architecture. Recently, researchers have identified the potential to apply distributed artificial intelligence (DAI) in solving various complex manufacturing system problems with software agents. In this regard, a multi-agent systems (MAS) is designed to decentralize the control of the manufacturing system, so as to reduce the complexity and to increase the flexibility of the manufacturing system. A typical MAS-based manufacturing system is composed of autonomous components (machines, tools, parts, etc.) which communicate with other components and perform their functions without a central controller.

The agent-based architecture is highly supportive to a wide range of manufacturing planning activities including process planning, scheduling, manufacturing enterprise integration, and supply chain management. However, the responsive behaviour of the MAS is usually ignored in most of the agent-based applications. Because of the dynamic nature of most production environments, as soon as an original plan or schedule is released to the shop floor, it will be inevitably and immediately subject to unexpected disturbances which may upset the execution. In consideration of the practical implementation of the agent-based concepts in real-time manufacturing environment, it is necessary for the MAS-based manufacturing planning and control environment to be equipped with dynamic control and quick response capabilities.

For instance, with regard to the development of MASs in manufacturing control, there are still no standard architectures for the agents and MASs. Manufacturing system designers have to design the agents and the MAS from scratch. A lot of effort has to be spent in building new structures. As a result, existing agent structures in manufacturing are mostly proprietary. In particular, most manufacturing control agents have structures which were originally established for computing applications. They are either purely belief-based or behaviour-based. For practical applications in manufacturing planning and control, agents in the MAS-based control system should possess both attributes. That is, an agent must be capable of planning its own course of action and at the same time it has to control a physical device which may be a machine tool, a robot or a material handling equipment in real-time.

This paper presents a hybrid MAS framework for manufacturing planning and control activities. The hybrid framework incorporates both the decentralized negotiation and hierarchical control. For designing agents in the MAS, a hybrid-based agent structure is also introduced.

II. RELATED RESEARCH

Many researchers have attempted to develop agent-based architectures to support manufacturing activities. Since the late 1980s, a number of researchers have applied agent technology to perform production planning and control on the shopfloor level [1][2]. For MAS in manufacturing, it is natural to use autonomous agents to represent manufacturing resources such as products, parts, operations, workers, workstations, machines, tools, fixtures, or automated-guided vehicles to facilitate manufacturing resource planning, scheduling, and execution control [3].

Two well-known MAS manufacturing applications are of particular significance in demonstrating these benefits. MetaMorph [4][5] was proposed as an adaptive agent-based architecture for concurrent engineering. Intelligent agents are established to model manufacturing entities including part and machines. They could pursue individual goals while satisfying constraints in combining the product design and production management issues. The MetaMorph architecture also demonstrates how to integrate with the legacy systems for an extension of its functionalities. In the Autonomous Agents for the Rock Island Arsenal (AARIA) project, Parunak *et al.* [6] investigated large-scale resource allocation and system simulation using autonomous agents, in the context of factory

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scheduling. The project envisaged agent-oriented manufacturing entities (e.g. human users, machines, parts) interacting over a network as a manufacturing enterprise whose performance and functionality could exceed that of centrally controlled manufacturing systems.

With many successful and important researches in agent technology and MAS-based manufacturing applications, there is still a lack of an unified framework which support both the planning and control activities simultaneously. From an enterprise's management point of view, Shen & Norrie [7] discussed a number of motivations that justified the importance of intelligence and responsive behaviour of agent-based approach to manufacturing systems. However, these characteristics are particularly difficult and expensive to achieve under dynamic changes and unexpected events. MetaMorph II [5] is a successor to the MetaMorph project. Its objective is to integrate the manufacturing enterprise's activities of design, planning, scheduling, simulation, execution, and product distribution, with those of its suppliers, customers, and partners within a distributed intelligent open environment. Although much focus has been put on agent coordination mechanisms, there is still not enough consideration on how the agent architecture could react to changes in real time.

In this paper, a unified MAS framework is proposed to support both the planning and control activities. The framework is based on results of some of the authors' previous endeavours in MAS-based manufacturing applications. Tang & Wong [8] proposed a reactive MAS for manufacturing system control. A prototype agent-based flexible manufacturing cell is established in which the agents can carry out simple task allocation via agent negotiation. Wong et al. [9][10] have proposed a MAS-based decentralized negotiation for solving complicated integrated process planning and scheduling problems. They further enhance the approach to support dynamic process planning and rescheduling problem under unexpected disturbances [11].

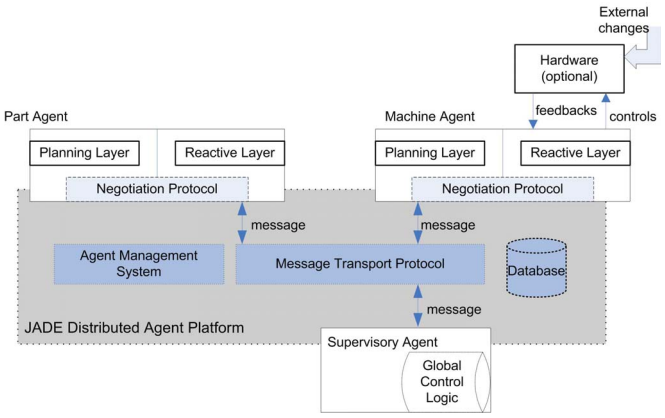


Fig. 1 Hybrid MAS framework.

III. HYBRID MAS FRAMEWORK

A hybrid MAS framework for manufacturing planning and control activities is proposed. Fig. 1 presents the MAS framework. It consists of local agents, a supervisory agent, and a distributed agent platform. The local agents include part agents (PA) and machine agents (MA) representing parts and machines respectively. All the planning and control decisions are made via a negotiation-based approach amongst the local agents. The Supervisory Agent (SA) is designed to coordinate the negotiation and monitor the local decisions made by the local agents. The agent platform mainly includes an agent management system, a message transport protocol, and a database. The agent management system controls the lifecycle of all the agents in the framework. The message transport protocol provides message passing mechanism between the agents. The database records the information flow and decisions made by the agents.

A. Hybrid-based negotiation

The hybrid-based negotiation mechanism is depicted in Fig. 2. It is characterized by the heterarchical negotiation between the local agents and the hierarchical control given by the SA.

For the heterarchical negotiation, the planning and control decisions are made between the local agents. In consideration of the product requirements and availability of manufacturing resources, the actual execution plan for producing a particular product will be established through negotiation between the PAs and MAs.

Since the local agents PAs and MAs in the simple heterarchical negotiation are self-interested, they achieve to optimize their own local objective functions only. It can be difficult for the agents to reach a final globally optimum solution. Hierarchical control is therefore established to overcome the problem. It mainly involves the addition of a the SA to the MAS architecture. The roles of the SA are twofold – it serves as a centralized controller and a system coordinator which provides hierarchical control and manages the negotiations between the local agents PAs and MAs respectively. The SA is empowered to access full information and status of all the agents in the system. Its primary function is to ensure that the global objective of the system is being observed. Depending on the computational ability of the SA and the system requirement, it is possible to consider one or several different global objectives such as minimizing jobs' makespan, minimizing jobs' tardiness, or balancing machines' loading etc. The timing and frequency of intervention are also determined by the SA.

A multi-agent negotiation protocol is necessary for

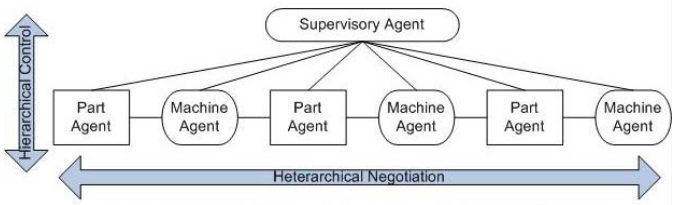


Fig. 2 Hybrid-based Agent Negotiation.

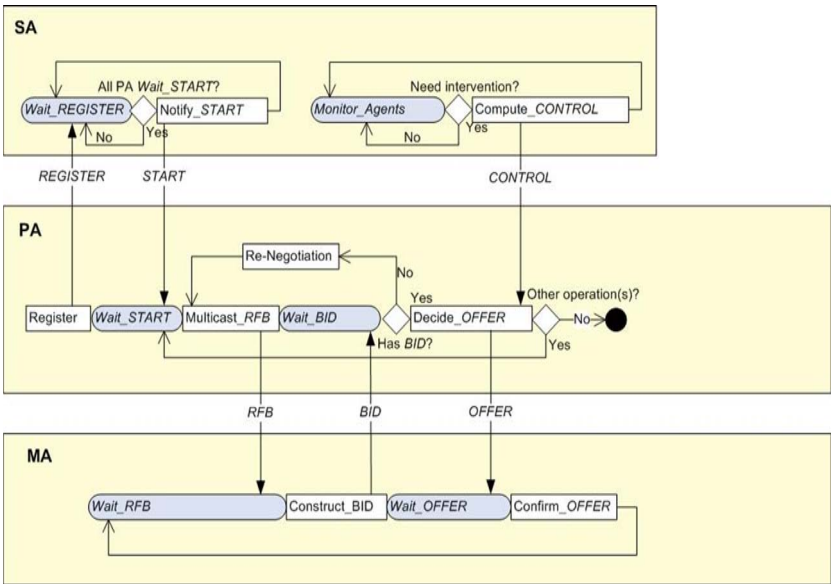


Fig. 3 - Hybrid Contract Net Protocol

effectively coordinating the interactions between the SA and the local agents. In this work, a hybrid contract net protocol (HCNP) is proposed to support both the heterarchical negotiation and the supervisory control. In addition, HCNP also supports multi-task many-to-many negotiations. Fig. 3 depicts a state transition diagram which explicitly shows the flow of operations of the agents. The agents' states and transitions are shown in the round-cornered boxes and sharp-cornered boxes respectively. The predefined communication message types are also listed. Note that all the states in the system are set with an expiration time to prevent infinite waiting of messages. HCNP has been successfully deployed to solve complicated IPPS problem [9][10][11]. The material-handling decisions are omitted for simplicity. If required, robot agents and material-handling agents can be established as in [8] to deal with assembly and transportation activities.

B. Hybrid Agent Structure

The internal structure of an intelligent agent usually comprises a communication unit, a decision unit, an operational unit, and a knowledge base (Fig. 4). These components represent different important features of an agent. Firstly, an agent should be able to interact with its environment or physical world via its operational unit. Secondly, an agent should be autonomous and be able to make decisions independently. An agent has to consider the status of environment and consult its knowledge base during decision making. The knowledge base stores the production rules and the "experience" of the agent and it defines the preference of the agent. Thirdly, a communication unit is included in the architecture. An agent should be able to interact with other agents in communication activities such as information exchange, command and feedback, etc.

As shown in Fig. 5, agents in the proposed MAS are established with a hybrid structure of two layers: a planning and a reactive control layer. The proposed agent structure is a combination of reactive (behaviour-based) and belief-based agents. Purely behaviour-based agents have quick response, but there is a lack of reasoning power for planning and execution functions. On the other hand, if purely belief-based agents are used, the agent will have reasoning power but the system will be slower in response and weaker in robustness. The proposed structure in this paper embraces the properties of these two distinctive types of agents in a hybrid layered approach.

The hybrid two-layer structure is a generic form of agents in the proposed MAS. The structure is compliant to the common structure shown in Fig. 4. Its reactive layer may be equivalent to the operational unit. This layer controls the hardware in the cell (robot and conveyor line, etc.) according to the decisions from the planning layer. The planning layer acts as the decision unit. It decides the appropriate actions during the manufacturing process. It makes decision by considering the machine status and consulting its knowledge base which is the database in the planning layer. Moreover, both planning and reactive layers have their own communication abilities.

In most cases, an agent contains the third layer of underlying hardware which is a physical entity that the agent controls. For machine control agent, the hardware may be a robot or a CNC machine. The reactive layer sends control signals to the hardware and gets feedback signal. On the other hand, an agent may not control any hardware and hence does not have any underlying hardware. For example, the supervisor agent that coordinates the operation of the whole system does not involve any hardware directly.

The reactive layer is responsible for the real-time operations of the agent. It employs the subsumption architecture and contains behaviours that couple stimuli and reactions directly. The layer is simple, has high effectiveness, and quick response. Its primary role is to ensure the execution of the plans and schedules. Due to the dynamic nature of the manufacturing system, the reactive layer provides real-time monitoring to the environments and to the hardware being connected. In the proposed generic structure, the reactive layer

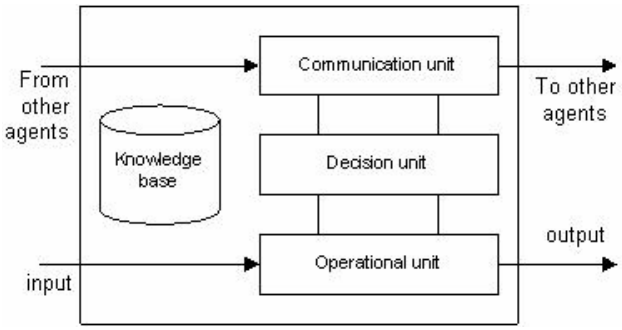


Fig. 4 Common structure of an agent.

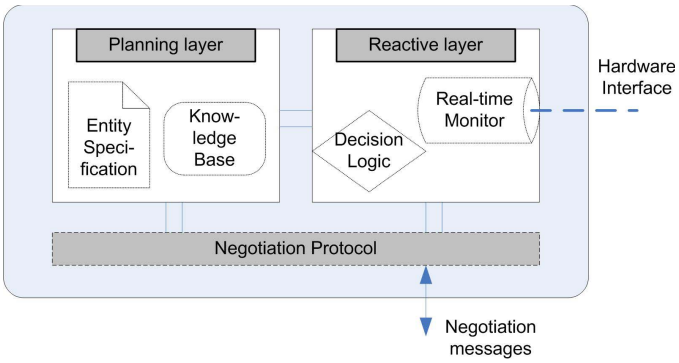


Fig.5 Hybrid layered structure of an agent

of one agent can send messages to the reactive layer of another agent. It greatly enhances the social ability of agents. It allows direct interactions between a number of reactive layers during real-time operation. Whenever a manufacturing disturbance occurs which may affect the original plans, it provides quick and real-time response to restore the plans and hence ensure the stability of the execution.

Unlike the reactive layer, the planning layer possesses all necessary reasoning power for schedule searching. It is designed to perform task allocation within the manufacturing systems. It contains specific information which states the agent's objectives and its own status. A knowledge base exists to provide historic information and decisions which aid the agent to make negotiation decisions internally. From primitive requirement data to a set of plans or schedules, the planning layer possesses all necessary reasoning power for making good negotiation decisions. During the planning stage, it communicates with other agents for planning via message passing. The planning layer also monitors and regulates decided schedules. It makes adjustments of schedules during the execution stage.

The planning layer and the reactive layer can communicate via messages. During the execution stage, the planning layer can send commands to the reactive layer and gets feedback. The planning and reactive layers are linked to each other so that the decisions and modifications can be updated instantly. They are both connected to the negotiation protocol (Fig. 3) which allows them to react and negotiate with other agents. In future, complex control problems may be broken down into simpler problems that can be handled individually by a single reactive layer. The planning layer will act as a coordinator of a number of reactive layers to solve control problems.

Furthermore, the proposed planning layer of the agent is hardware independent. If it is required to change any hardware, only the reactive layer needs to be modified or replaced. The overall structure will not be affected. Thus, the final design of the agent can be easily modified and used in agents with similar functions.

C. *Distributed agent platform*

The local agents, the supervisory agent, and the negotiation protocol are all implemented in the Java Agent Development environment JADE platform. It is a well-established platform for development agent applications and it is conformed to the FIPA (Foundation of Intelligent Physical Agent) standards. In JADE, user-defined agents are situated at agent containers. Containers may be in one or several computers that JADE are installed. Agents can communicate with others by passing agent-communication language (ACL) messages among or within containers. JADE is multi-platform and can be extended easily. Agents in the proposed system, including the SA, PA, and MA, can be easily extended from the basic agent class. Fig. 6 shows the implementation of the MAS framework onto three networked computers.

IV. MAS-BASED PLANNING AND CONTROL

To illustrate the functioning of the MAS-based planning and control environment, a simple manufacturing system with 4 machines and 2 part types is shown in Fig. 7.

Agents in the system are designed with the hybrid approach that they contain planning and reactive control layers. Each CNC machine is controlled by a machine agent (MA) which monitors the machine and instructs the machine tool to carry out the machining tasks according to a predefined schedule. Each part type is represented by its corresponding part agent (PA) which provides the required part design and processing details. A supervisory agent (SA) acts as the supervisor and coordinator of the other subordinate agents.

In the proposed system, manufacturing orders are processed through two stages: planning and execution. The planning layers of the agents are involved in the planning stage. The reactive layers are responsible for the execution stage.

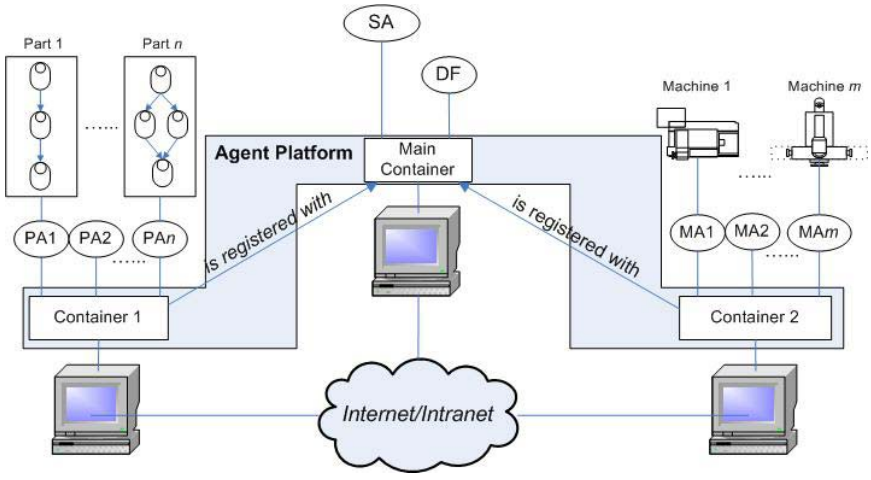


Fig. 6 Implementation of the MAS framework.

A. *Integrated Process Planning and Scheduling*

The function of the planning stage is to determine the actual manufacturing process plan and the schedule within the manufacturing system. The authors have engaged in the development of agent-based integrated process planning and scheduling (IPPS) systems [9][10][11]. A MAS-based decentralized negotiation has been established for solving the IPPS problems, with advantages of flexible system architectures and responsive fault tolerance. In consideration of the scheduling requirements and availability of manufacturing resources, the actual process plan and schedule for producing a particular product are determined through negotiation between PAs and MAs and the hierarchical control of the SA under the hybrid MAS framework.

For each part in the current IPPS system, all feasible manufacturing processes and routings are recorded as alternative process plans. Similarly, alternative machines for an operation are also included. The complete set of feasible production processes and sequences for each part are recorded in an AND/OR graph. Fig. 7 shows an IPPS problem involving 2 parts and 4 machines. The two example parts are extracted from Wong *et al.* [12] and Zhang *et al.* [13].

Under the hybrid negotiation protocol in Fig. 3, bids are evaluated in accordance with a currency function which considers an agent's multi-objectives and IPPS parameters. The currency function, which incorporates the important IPPS parameters – processing time and due date, is adopted for the preparation and evaluation of bidding prices. For the decentralized negotiation amongst the local agents, a PA will invite eligible MAs to bid for immediate available operations from its flexible process plan. An IPPS problem is completely solved when all the PAs have determined the suitable

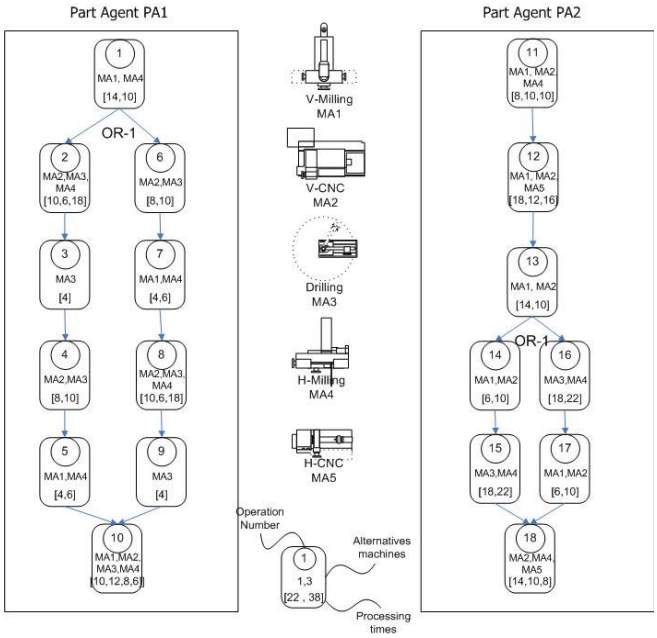


Fig.7 Alternative process plans of the two parts.

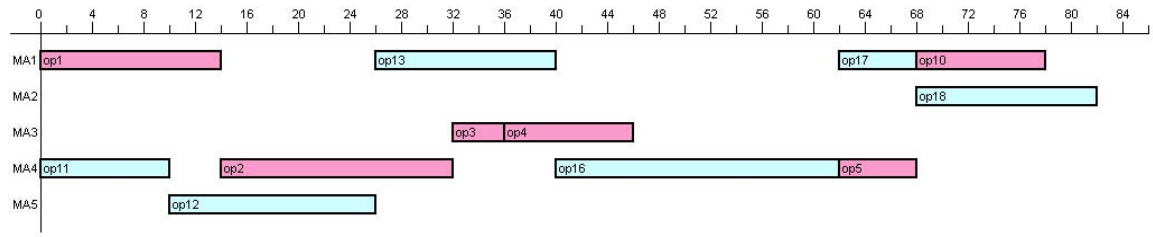


Fig.8 Negotiation result of the two example parts.

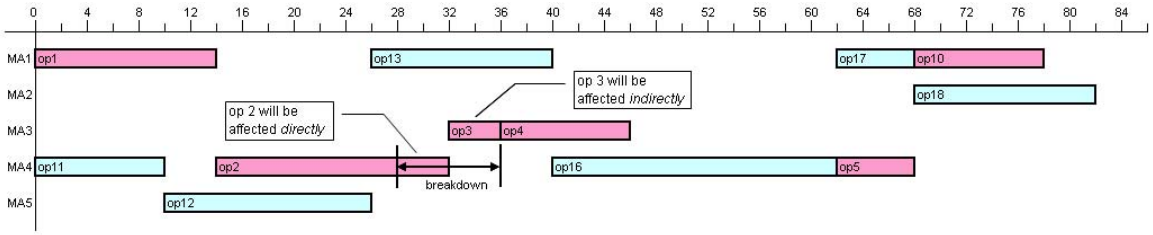


Fig.9 An example of machine breakdown and the affected operations.

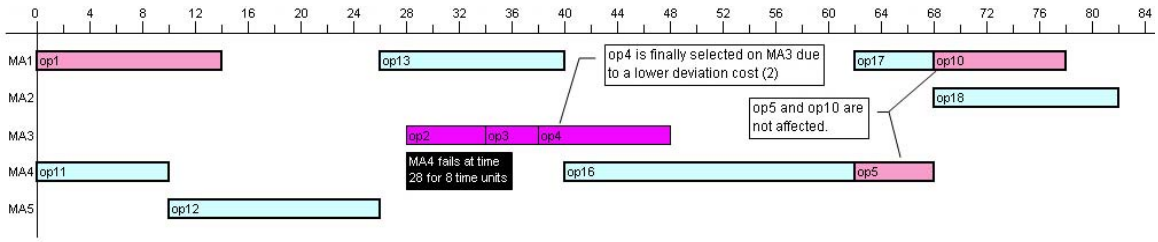


Fig.10 Rescheduling result.

machines and schedules for all the required operations. The SA monitors the negotiation process and keeps track of the performance of the system. All the decisions made by the PAs are monitored by the SA. Nevertheless, the SA will only intervene if there is a conflict between the local decisions and the system objective. The global decisions made by the SA will be directly imposed to the PAs and the SA will monitor the agents again. The characteristics and detailed implementation of the hybrid control scheme for the IPPS problems including the formulation of the flexible process plans, the currency function, the agents' local decision models, and the global control of the SA are available in [9], [10] and [11]. The negotiation result of the two example parts is presented in Fig. 8.

B. Dynamic process planning and rescheduling

Once a part is loaded to the manufacturing environment, it is inevitably subjected to unexpected production disturbances such as machine breakdowns, rush order arrivals and order cancellations. A process plan may become inefficient or even infeasible due to changes. These uncertainties cause deviations to the original schedules and will finally affect the performance of the manufacturing systems. To cope with the unexpected disturbances, it is necessary to revise the process plan and the schedule dynamically.

The proposed hybrid MAS framework is tailored to support online manufacturing control. The reactive layer and the negotiation protocol of the internal structure of the agents have been designed to support dynamic process planning and rescheduling. In this research, the most common disturbances in job shops, machine breakdown, is considered. Its impacts to the schedule and the performance of manufacturing systems are examined. As inspired by the Affected Operations Algorithms [14][15], it is important to identify those operations which are affected by the disturbances, either directly or indirectly. These affected operations will then be rescheduled.

A case of machine breakdown is presented in Fig. 9. The breakdown of machine MA4 occurs at time 28 for a duration of 8 time units. Since operation op2 was originally scheduled within that duration, it will be directly affected and hence it is added into the set of affected operations AO, thereby AO={op2}. As shown in the AND/OR graph (Fig. 7), op2 can be processed with alternative machines MA2, MA3, or MA4, with processing times 10, 6, 18 time units respectively. PA1 can determine that the expected earliest finishing time of the new schedule of op2 equals 28+6=34 by MA3. Since the delayed finishing time is lagging the pre-scheduled starting time of op3 (32), op3 will be affected indirectly and it will be added to AO, hence AO={op2, op3}. The result of rescheduling is listed in Fig. 10.

V. CONCLUDING REMARKS

A hybrid MAS framework for manufacturing planning and control activities is presented. It is featured with a hybrid negotiation-based approach for distributed task allocation. The framework supports both the heterarchical negotiation and hierarchical control. A generic agent structure is proposed which incorporates both the planning and reactive functions of an agent. A hybrid negotiation protocol has been designed to support the hybrid control scheme.

The MAS framework has been implemented in a multiagent manufacturing system for integrating process planning and scheduling. With the hybrid approach, the global vision and control provided by the supervisory agent has an essential effect over the decentralized negotiation.

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